

Unified Spacetime Substrate Configuration

Theory:

A Unified Cosmological Model Based on Continuum Medium Phase Transition

Abstract

This paper establishes an axiomatic Spacetime Phase State Theory (SPT) grounded in the ontology of global continuous spacetime, without assuming any predefined microscopic spacetime structures. By extending the configurational gravitational description derived from conventional scalar-tensor gravitational theory, the paper interprets pivotal issues in cosmology and fundamental physics. The theory strictly complies with Lorentz covariance, maintains mathematical self-consistency with General Relativity, and agrees with the Planck 2018 cosmological parameter measurements.

Adhering to the monistic ontological principle, the theory excludes hypothetical dark matter particles, independent dark energy entities, diverse fundamental quantum fields and any extraneous substances beyond spacetime. It proposes that all observable physical phenomena manifest as extrinsic behaviors stemming from configurational evolution and local constraint responses of a single continuous substrate.

No rigid definition is imposed on the inherent primordial state of the substrate; deformation and phase transition serve as core evolutionary rules. The unified substrate undergoes progressive phase transitions: collapsed configurations correspond to dark matter components, while potential effects of stretched configurations manifest as dark energy. The physical origin of Modified Newtonian Dynamics (MOND) is identified as the first-order phase transition critical point of the substrate, and Hubble tension arises from spatial domain phase transitions of the substrate during cosmic evolution.

The theory clarifies hierarchical compatibility between macroscopic substrate evolution and microscopic particle physics. Established laws of particle physics and quantum mechanics remain fully applicable at microscopic scales, whereas macroscopic gravitational behaviors and cosmic evolution are governed by substrate configurational mechanisms. Endowed with intrinsic entropy-increasing properties, the substrate evolves irreversibly from an ordered state toward disorder. This framework coherently resolves fundamental puzzles including gravitational essence, quantum nonlocality and antimatter gravitational symmetry. Multiple falsifiable predictions are put forward with complete verification value, offering a consistent and rational new research

direction to break through the limitations of the Λ CDM paradigm.

Keywords: Unified Spacetime Substrate; Configurational Scalar-Tensor Gravity; Dark Matter; Modified Newtonian Dynamics (MOND); Hubble Tension; Falsifiability; Spacetime Entropy Increase; Hierarchical Physical System

1 Introduction

Modern cosmology and fundamental physics face persistent longstanding challenges. Decades of dark matter particle detection have yielded no substantial breakthroughs, research on the unification of four fundamental forces has stagnated, Hubble tension has expanded to a 5σ confidence level, and the widely validated MOND framework at galactic scales still lacks a credible physical origin.

The mainstream Λ CDM model relies on numerous unconfirmed assumptions and constantly introduces extra parameters such as early dark energy and sterile neutrinos to reconcile observational discrepancies, violating Occam's razor principle.

This paper presents a monistic cosmological model based on continuum medium phase transition. No entities outside spacetime are postulated, and all physical phenomena originate from configurational evolution of the unified substrate. Dark matter and dark energy are not independent cosmic components, but phase manifestations of the identical substrate across different evolutionary stages and spatial regions. Gravity emerges macroscopically from overall substrate configurational variations, belonging to a distinct dynamic hierarchy separate from electromagnetic, strong and weak interactions.

A hierarchically compatible structure is adopted. Established experimental-verified particle physics and quantum mechanics govern microscopic particle interactions without revision, while macroscopic spacetime and gravitational evolution are dominated by substrate configurational rules. The substrate undergoes irreversible entropy-driven evolution, forming the fundamental logic of unidirectional cosmic development. This theory eliminates inherent logical flaws of conventional models and achieves unified interpretation of core physical problems across multiple fields.

2 Core Theoretical Interpretation

2.1 Hypothesis of Unified Continuous Substrate

The universe possesses a sole primordial continuous unified substrate, with no separable internal structure detected within current observational scales. It transcends the classical dichotomy of matter and energy, acting as the common carrier of spacetime, conventional matter, dark matter and all fundamental interactions.

Matter is embedded entirely within the substrate rather than existing independently, analogous to pulp embedded in jelly; no free matter can exist detached from the substrate. The theory follows state neutrality without

predetermining the substrate's initial solid, liquid or gaseous form. Macroscopic states of conventional matter are merely figurative analogies and not theoretical criteria. All deductions are based on dynamic laws of substrate deformation and phase transition, whose mechanisms and gravitational logic remain unaffected by initial states.

Dark matter is defined as collapsed configurations formed by substrate evolution, excluding independent microscopic particles and classical ether-like media.

A hierarchical physical architecture is established: microscopic phenomena including fundamental particle composition, interactions and quantum reactions fully abide by validated particle physics and quantum mechanics without new microscopic rules. Macroscopic spacetime evolution, gravitational effects and large-scale cosmic structure formation are governed by intrinsic substrate configurational evolution. The two systems operate coherently without conflict.

The substrate maintains extreme stability. Ordinary physical processes at conventional energy scales only trigger minor local deformation without altering essential substrate properties. Profound substrate configuration perturbations can only occur under extreme ultra-high energy conditions, incapable of reversing the overall evolutionary trend.

2.2 Configuration Variation Mechanism

Intrinsic attributes of the substrate remain invariant throughout evolution, with state changes limited to deformation and phase transition. Deformation refers to gradual smooth variations including compression, curvature and density rearrangement induced by external effects. Phase transition denotes abrupt structural reconstruction once accumulated deformation reaches the critical threshold. Both types of variation reshape extrinsic morphology rather than generating new primordial entities.

2.3 Spacetime Positioning

Spacetime constitutes the fundamental macroscopic configuration formed by fully stretched substrate, serving as an integral part of the substrate system instead of the ultimate cosmic ontology. All evolutionary characteristics such as spacetime curvature and cosmic expansion are directly determined by substrate configurational variations.

3 Core Axiom System (Fundamental Postulates)

1. The substrate possesses inherent elasticity and undergoes continuous global or local deformation.
2. An intrinsic critical deformation threshold exists; local deformation accumulation beyond this value may trigger structural phase transition.
3. The substrate features inherent entropy-increasing evolution. The universe evolves irreversibly from order to disorder, maintaining dynamic equilibrium with elastic potential energy and dominating global

macroscopic cosmic evolution. Local matter can form ordered structures and partial entropy reduction, yet cannot reverse global entropy growth.

4. Weak coupling autonomous evolution hypothesis: Dynamic evolution and spontaneous phase transition of the unified substrate are predominantly driven by inherent elastic potential energy and entropy growth trends. At typical astrophysical scales, three-dimensional matter distribution and spacetime curvature exert negligible back-coupling effects, only generating weak local constraints with marginal influence on global evolution and large-scale phase transition. Observable remarkable back-reaction effects merely emerge in extreme high-energy environments or vicinity of supermassive compact celestial bodies.

4 Core Physical Logic (Inferences Derived from Axioms)

4.1 Hypothesis on Gravitational Origin

Local substrate deformation gradients directly generate spatial curvature and gravitational gradients, constituting the fundamental source of gravity. Celestial bodies and microscopic particles passively gather within gravitational potential wells formed by substrate deformation, rather than mass governing gravitational generation.

Gravity is a macroscopic emergent effect of global substrate configuration shifts, while electromagnetic, strong and weak interactions are local constraint effects. Their dynamic essences differ fundamentally, making unification of four fundamental forces unattainable within the existing framework. Microscopic particle interactions follow established quantum and particle physics laws, operating separately from macroscopic substrate gravitational evolution.

4.2 Deformation-Phase Transition Critical Mechanism

Persistent elastic deformation accumulates within the substrate. When local deformation gradients reach the universal characteristic threshold $a_0 \approx 1.2 \times 10^{-10} \text{m/s}^2$, first-order phase transition readily occurs, transforming stretched substrate into collapsed morphology. Determined by the balance between substrate elastic potential energy and cosmic entropy growth, the threshold is an intrinsic substrate property rather than a fixed constant, fluctuating slightly with local constraint intensity and spatial energy density.

4.3 Potential Origin of MOND Phenomenological Law

The critical acceleration observed in MOND corresponds to the phase transition threshold between stretched and collapsed substrate states. When local gravitational acceleration far exceeds the characteristic threshold, local material constraints suppress collapsed configuration formation, and gravitation conforms to Newtonian mechanics and General Relativity. When acceleration drops substantially below the threshold, spontaneous substrate phase transition produces collapsed structures and additional gravitational effects, matching low-acceleration observational behaviors of MOND.

Local matter inhibits substrate phase transition, enabling deduction of the baryonic Tully-Fisher relation. Phase transition thresholds rise synchronously

with spatial constraint intensity; higher temperature and density of intracluster gas elevate critical values, explaining MOND's inapplicability at galaxy cluster scales.

4.4 Potential Explanation for Hubble Tension

During the early cosmic photon decoupling era, large-scale substrate phase transition was absent, and the uniformly stretched substrate drove cosmic expansion at a velocity corresponding to the Hubble constant $H_0 \approx 67.4 \text{ km/s/Mpc}$ measured via cosmic microwave background radiation.

Progressive regional domain phase transitions increased the proportion of collapsed substrate configurations as the universe evolved. Intrinsic potential energy of collapsed structures elevates local expansion rates, consistent with the higher Hubble constant $H_0 \approx 73.0 \text{ km/s/Mpc}$ derived from nearby celestial observations. Hubble tension originates from staged substrate phase transitions and uneven spatial phase distribution, requiring no additional cosmic components for interpretation.

4.5 Unified Framework for Fundamental Physical Puzzles

- Quantum Nonlocality: The unified substrate maintains global connectivity. Quantum entanglement constitutes instantaneous responses to holistic substrate constraints without superluminal signal transmission, complying with the principle of constant light speed. Established quantum theories remain valid, with macroscopic substrate connectivity supplementing physical origins.
- Antimatter Gravitational Symmetry: Gravity responds solely to overall substrate energy density independent of material phase. Matter and antimatter generate equivalent spacetime curvature and mutual attraction, consistent with antihydrogen gravitational measurement results. Microscopic interactions between matter and antimatter follow conventional particle physics rules.
- Cosmic Expansion Mechanism: Cosmic expansion proceeds without hypothetical dark energy entities, jointly propelled by natural substrate stretching, energy dissipation of collapsed configurations and cosmic boundary effects. Both uniform and inhomogeneous expansion can be coherently interpreted, aligning with irreversible global entropy growth of the substrate.

4.6 Fundamental Logic of Cosmic Evolution

The substrate evolves unidirectionally and irreversibly toward entropy growth and stretching, manifested as continuous macroscopic cosmic expansion. Cumulative local deformation triggers successive multi-stage phase transitions, forming gravitational frameworks, conventional matter and compact celestial bodies. Global entropy growth is an inherent substrate property unaffected by local material motion and celestial evolution. Local ordered structures and partial entropy reduction are secondary derived phenomena incapable of altering overall cosmic evolutionary direction. Cosmic expansion, galactic motion, light propagation, fundamental interactions and quantum

phenomena all represent extrinsic physical characteristics of the unified substrate under varying deformation degrees and configurational states.

5 Correspondence Between Four Steady-State Substrate Configurations and Cosmic Phenomena

Note: Stretched, collapsed, compact and extreme states refer exclusively to intrinsic substrate configurations, distinct from classical solid, liquid and gaseous states of ordinary matter. Condensed properties of daily substances are secondary derivatives of compact substrate configurations, irrelevant to primordial substrate morphology.

1. Stretched Configuration: Mild substrate deformation below phase transition criteria forms macroscopically flat spacetime, dominating cosmic void regions and generating dark energy-driven expansion effects.
2. Collapsed Configuration: Deformation accumulation exceeding critical values induces first-order phase transition, forming dark matter gravitational halos and large-scale cosmic gravitational frameworks.
3. Compact Configuration: Further deep deformation of collapsed structures triggers second-order phase transition, evolving into microscopic particles and celestial bodies. Strong local constraints hinder secondary collapse, and gravitational behaviors follow General Relativity and Newtonian mechanics. All microscopic particle properties and interactions conform to standard particle physics.
4. Extreme Configuration: Ultimate substrate collapse triggers terminal phase transition, corresponding to cosmic black holes.

6 Theoretical Boundary Definition

6.1 Consistency with Existing Observations

1. Mathematically self-consistent with General Relativity and fully satisfying Lorentz covariance.
2. Reproducing galactic-scale observational outcomes of MOND and deriving the baryonic Tully-Fisher relation.
3. Compatible with all parameters from Planck 2018 cosmic observations.
4. Predictions on antimatter gravitational properties consistent with experimental measurements.
5. Fully compatible with verified microscopic physical conclusions without overturning established particle physics and quantum mechanics laws, only reconstructing fundamental mechanisms of macroscopic spacetime gravity and cosmic evolution.

6.2 Extended Speculations (Pending High-precision Observational Verification)

1. Four fundamental interactions and diverse quantum effects essentially arise from differentiated microscopic substrate deformation

modes.

2. Light speed is not an absolute constant, varying dynamically with substrate density and deformation gradients.

3. Intrinsic substrate features can be indirectly inferred via precise gravitational wave detection and dark matter halo mapping.

7 Testable Predictions and Verification Approaches

Falsifiable predictions are proposed for validation via current and future astronomical observations and physical experiments:

1. Dark Matter Particle Detection: Underground capture and collider experiments based on conventional particle detection principles will fail to identify primordial dark matter particles. Discovery of dark matter particles would falsify core theoretical hypotheses.

2. Critical Threshold Characteristics: Galactic gravitational acceleration features the demarcation threshold $a_0 \approx 1.2 \times 10^{-10} \text{m/s}^2$, with classical Newtonian gravity breaking down below this value. Galaxy cluster thresholds are notably higher and positively correlated with intracluster gas temperature.

3. Isolated Dark Celestial Bodies: No standalone dark matter objects detached from baryonic matter exist, nor do ordinary galaxies completely devoid of dark matter halos.

4. Gravitational Structure Formation Timeline: Dark matter halos form considerably earlier than stars and planets; supermassive black holes may emerge earlier than predicted by mainstream theories.

5. Absolutely Flat Spacetime: Weak deformation gradients permeate all cosmic regions, with approximate flatness resulting from subcritical deformation levels.

6. Configuration-dependent Local Light Speed: Light speed declines in regions with intense gravity and severe substrate collapse, reaching maximum values in empty stretched cosmic space.

7. Gravitational Wave Propagation Time Difference: Gravitational waves arrive marginally earlier than electromagnetic waves from identical cosmic events.

8. Gravitational Wave Dispersion: Gravitational waves of different frequencies travel at slightly varied velocities across dark matter phases, with high-frequency components propagating slower than low-frequency counterparts.

9. Black Hole Central Structure: No mathematical singularity with infinite density exists at black hole centers, which consist of finitely dense extreme collapsed substrate. Ringdown waveforms of merging black holes deviate from singularity model calculations.

10. Galaxy Cluster Collision Features: Spatial offsets between gravitational lensing regions and visible matter shock fronts in systems such as the Bullet Cluster stem from substrate phase reconstruction

rather than particle interactions.

11. CMB Power Spectrum Features: Early spacetime phase transitions leave distinctive suppression signatures on large-scale cosmic microwave background angular power spectra.

8 Core Theoretical Advantages

1. Compliance with Occam's Razor: Deductions rely solely on a single primordial substrate and basic rules of elastic deformation and critical phase transition, avoiding unconfirmed dark matter particle and dark energy hypotheses and remedial extra parameters.

2. Strong Openness and Universality: No arbitrary restrictions on initial substrate states, with core evolutionary logic independent of primordial morphology.

3. High System Compatibility: Macro-micro hierarchical architecture fully accommodates established particle physics and quantum mechanics without revamping experimentally validated microscopic laws, innovating only macroscopic cosmology and gravitational fundamentals.

4. Powerful Unified Interpretation: Coherently explaining gravity essence, dark matter, dark energy, cosmic expansion, galactic evolution, black hole properties, quantum effects and antimatter gravity, complementing fundamental unidirectional cosmic evolution mechanisms.

5. Strict Falsifiability: All core inferences equipped with corresponding observational criteria for empirical verification.

6. High Mathematical Compatibility: Retaining mature computational frameworks of General Relativity and continuum mechanics, compatible with existing galactic gravity and cosmological numerical models.

9 Preliminary Mathematical Implementation Path

1. Abandon microscopic particle statistics and thermodynamic calculations, adopting substrate deformation density, deformation gradient and phase transition threshold a_0 as core computational variables.

2. Utilize continuum mechanics, General Relativity deformation field equations and Configurational Scalar-Tensor Gravity (CSTG) mathematical structures, removing unvalidated particle coupling terms and direct scalar field-Ricci scalar coupling terms while reserving only substrate deformation evolution terms.

3. Classify mathematical boundaries of four configurations via the phase transition criterion a_0 , introducing environmental correction factors to characterize threshold fluctuations for galactic gravity and cosmic expansion numerical simulation.

4. Conduct theoretical deduction and verification within existing astronomical gravitational computational frameworks without constructing entirely new complex mathematical systems.

10 Theoretical Verification System

Self-consistent computational logic is achieved. Subsequent verification requires no overhaul of existing calculation frameworks. Empirical testing proceeds by distinguishing two physical interpretations — unified substrate configuration and dark matter particles — through astronomical and experimental observations.

10.1 Mandatory Core Verification Items

1. Universality Test of Critical Acceleration Threshold: Verify whether gravity deviates from Newtonian law around $a_0 \approx 1.2 \times 10^{-10} \text{m/s}^2$ across galaxies, dwarf galaxies, clusters and cosmic void edges, and measure threshold variations correlated with local constraints and energy density.
2. Gravitational Structure Chronology Test: Confirm dark matter halo formation precedes stellar and baryonic matter emergence via James Webb Space Telescope observations of early cosmic gravitational potential wells.
3. Search for Isolated Dark Matter Objects: Conduct all-sky gravitational lensing surveys to examine standalone dark matter halos unassociated with ordinary matter.
4. Black Hole Internal Structure Test: Detect measurable discrepancies between observed black hole merger ringdown waveforms and singularity model predictions using LIGO and future space-based gravitational wave detectors.

10.2 Auxiliary In-depth Verification Items

1. Local Variable Light Speed Test: Measure light speed differences between dark matter halo cores, black hole vicinities and empty flat cosmic space through high-precision astronomical ranging and pulsar timing.
2. Gravitational and Electromagnetic Wave Arrival Time Test: Identify systematic gravitational wave lead time in multi-messenger observations of neutron star mergers and gamma-ray bursts.
3. Dark Matter Halo Structure Test: Validate continuous medium distribution characteristics without discrete particle clustering via strong gravitational lensing fine mapping.
4. Coupling Test of Cosmic Expansion and Substrate Stretching: Correlate cosmic expansion acceleration with large-scale stretched substrate density distribution through supernova surveys and cosmic structure mapping.
5. Galaxy Cluster Collision Characteristic Test: Further confirm phase reconstruction-induced spatial offsets via X-ray and high-

precision gravitational lensing observations of colliding galaxy clusters.

10.3 Exclusive Supporting Evidence

Persistent absence of dark matter particle signals from global underground detectors and high-energy colliders serves as exclusive corroborative evidence supporting the unified substrate theory.